

NAME:

STUDENT ID:

1. What type of variable is being measured by each of the following questions?

**A**

How much coding experience do you have?

- ☐ None.
- ☐ Very little. I've just dipped my toe in.
- ☐ A bit. I've used coding in a limited capacity, such as for a small part of a course
- ☐ Some. I've had to write code for one or two classes and am comfortable with it.
- ☐ A good deal. I have several years of experience writing code.

**B**

What year are you in your studies?

- ☐ 1st year / freshman
- ☐ 2nd year / sophomore
- ☐ 3rd year / junior
- ☐ 4th+ year / senior

**C**

Please also rate that experience on a scale from 1 to 10.

- 1 2 3 4 5 6 7 8 9 10
- No experience ☐ ☐ ☐ ☐ ☐ ☐ ☐ ☐ ☐ ☐ Tons of experience

**D**

Are hotdogs sandwiches?

- ☐ Yes
- ☐ No

**E**

What zip code do you consider home? (If from abroad, use UC Berkeley's 94720)

2. Last class you collected data on six different variables on your classmates. List the data types of each of those variables, the units that they were measured in (if it's continuous numerical), and the range of possible values that you would expect them to take.

- Name:
- Contact info:
- Hometown:
- # Siblings:
- # Semesters in college:
- Farthest distance from campus during the break (assume it's in miles)

3. As discussed in the lecture notes, this is **not** a data frame:

|                  |       | Alien color |      |       |
|------------------|-------|-------------|------|-------|
| Time of sighting |       | Green       | Blue | Total |
|                  | Night | 55          | 11   | 66    |
|                  | Day   | 23          | 55   | 78    |
|                  | Total | 78          | 66   | 144   |

But **where did this data come from?** Imagine you are researchers who are collecting this data, and you record your observations as rows in a dataframe.

- What would the *unit* of observation be?
- How many rows would there be? How many columns would there be?
- Sketch a schematic of the data frame based off of your answers to the above (just column names and a few example rows)

@. Pull out your worksheet from yesterday's class again! You'll now be storing some of the data you collected using R code.

- Note: if you haven't already, you really need to read today's tutorial (to learn to code) and watch the video about R Studio (so you know how to use our coding environment).
  - Open R Studio, create a new Quarto document, and save it.
  - Insert a new code cell.
  - Create three vectors: name, hometown, and sibs\_and\_pets filled with your data.
  - **You don't need to turn in the code file itself** - just copy down the code you wrote, once it's working.
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- Combine the three vectors into a data frame called my\_classmates. Copy the code you used below.